

Chapter -6

- **Web Content Mining** is the process of extracting useful information from the contents of web pages, such as text, images, audio, video, and structured data like tables or metadata.
- It helps in discovering patterns, trends, and insights from web content using techniques from data mining, natural language processing, and machine learning.

- A web crawler (also called a spider or bot) is a software program that automatically browses the internet to collect information from web pages.
- In simple words:
- A web crawler visits websites, reads their content, and stores data (like text, links, or images) for later use — usually by search engines or data mining tools.
- Example:
- When you search something on Google, the results appear because Google's web crawler (Googlebot) has already visited and indexed millions of web pages.
- Main purposes:
- To discover new or updated web pages
- To collect data for web mining or research
- To index pages for search engines so users can find them easily

Role of Web Crawlers in Web Mining:

- Web crawlers, also known as spiders or bots, play a crucial role in web mining by automatically browsing and collecting data from web pages. They systematically explore the internet, download web content, and store it in a central database for analysis.
- Their main functions include:
- **Data Collection:** Gathering web pages and updating information from websites.
- **Indexing:** Organizing the collected data so that it can be efficiently searched and mined.
- **Filtering:** Identifying and storing only relevant or new information.
- In short, web crawlers act as the foundation of web mining by providing the raw data needed for content, structure, and usage analysis.

- PageRank measures the **importance of a web page** based on the number and quality of links pointing to it. A page is considered more important if it is linked by many other important pages.
- **How it works:**
 - Each web page starts with an equal rank.
 - When a page links to another, it “**votes**” for that page.
 - Pages that receive more votes (links) from high-ranking pages get **higher PageRank scores**.
- **In short:**
 - PageRank helps search engines decide **which pages should appear first** in search results, ensuring that more trustworthy and relevant pages rank higher.

- A page is considered **important** if other **important pages link to it**.
- ⚙️ **Basic Idea**
- Every webpage gets a score called **PageRank (PR)**.
- If page A has a link to page B, it gives part of its importance to page B.
- The more links a page receives from other high-ranking pages, the higher its PageRank becomes.

PageRank Formula

$$PR(A) = (1 - d) + d \times \sum \frac{PR(B)}{L(B)}$$

Where:

- **PR(A)** = PageRank of page A
- **d** = Damping factor (usually 0.85) — it represents the chance that a user continues clicking links
- **PR(B)** = PageRank of page B (pages that link to A)
- **L(B)** = Number of outbound links from page B
- The summation is over all pages **B** that link to **A**

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- **🔍 Steps in PageRank Calculation**
 1. **Assign initial ranks** to all pages equally (e.g., if 4 pages, each gets $1/4 = 0.25$).
 2. **Calculate new PR values** using the formula for each page.
 3. **Repeat (iterate)** until the PR values stabilize (converge).
 4. **Rank pages** — the higher the PageRank, the more important the page.

- **Example**
- Consider 3 web pages: **A, B, and C**
- **Links:**
- $A \rightarrow B, C$
- $B \rightarrow C$
- $C \rightarrow A$
- Damping factor, **$d = 0.85$**

Step 1: Initial ranks

Each page = $1/3 = 0.333$

Step 2: Apply formula

$$PR(A) = (1 - 0.85) + 0.85 \times [PR(C)/L(C)]$$

$$PR(B) = (1 - 0.85) + 0.85 \times [PR(A)/L(A)]$$

$$PR(C) = (1 - 0.85) + 0.85 \times [PR(A)/L(A) + PR(B)/L(B)]$$

After several iterations, the approximate values become:

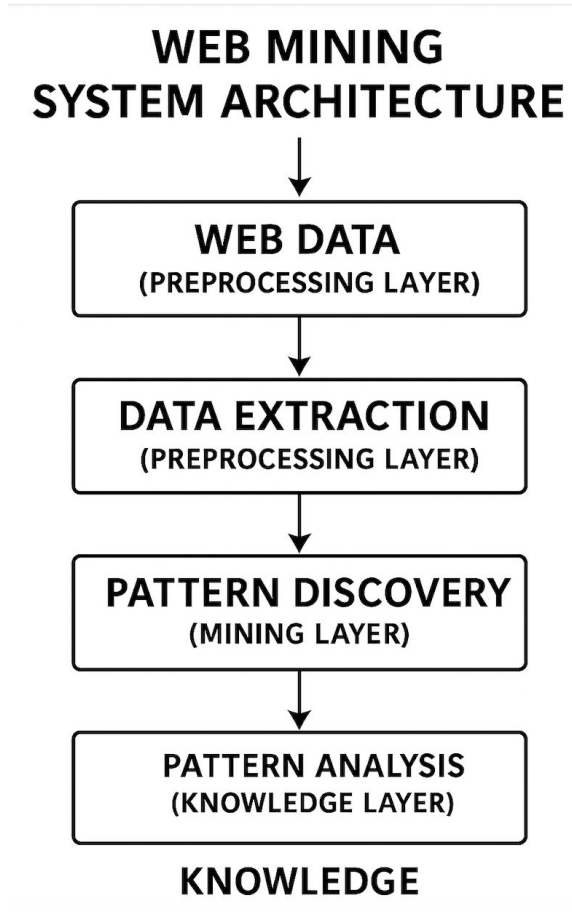
Page	PageRank
A	0.384
B	0.213
C	0.403

So, **Page C** has the highest rank — it's the most important page in this network.

- PageRank depends on both **quantity and quality** of links.
- A link from a **popular (high PR)** page carries more weight.
- It helps **search engines rank results** — higher PageRank pages usually appear higher in search results.
- It prevents spam by valuing **reliable, trusted links** more than random ones.

Feature	Web Content Mining	Web Structure Mining
Definition	Extracts useful information from the actual contents of web pages.	Analyzes the structure of links between web pages.
Focus Area	Text, images, audio, video, and other data available on web pages.	Hyperlinks, connections, and relationships among web pages.
Objective	To understand and retrieve meaningful data from web content.	To identify the structure, hierarchy, and importance of web pages.
Techniques Used	Text mining, natural language processing (NLP), machine learning.	Graph theory, link analysis algorithms (like PageRank, HITS).
Example	Extracting customer reviews, product descriptions, or news articles.	Analyzing how many and which websites link to a particular page.
Output	Patterns, keywords, summaries, or insights from web data.	Link-based ranking or web graph structure.

Web Mining Architecture



- **Web Mining System Architecture**
- A **Web Mining System** extracts useful knowledge or patterns from web data (such as web content, structure, or usage). Its architecture generally has **three main components**:
-  **1. Data Extraction (Preprocessing Layer)**
- Collects and prepares raw web data from different sources.
- Uses **web crawlers**, **parsers**, and **log analyzers** to gather data.
- Cleans and converts unstructured web data into a structured format suitable for mining.
-  **Example:**
Extracting HTML content, removing tags, and identifying keywords or links.
-  **2. Pattern Discovery (Mining Layer)**
- Applies data mining techniques to discover useful patterns or relationships.
- Techniques include **clustering**, **classification**, **association rules**, and **sequence analysis**.
- Depending on the type of mining:
 - **Web Content Mining** – analyzes text, images, etc.
 - **Web Structure Mining** – studies hyperlinks and connectivity.
 - **Web Usage Mining** – examines user behavior from logs.

- **Example:**
Finding that users who visit “product A” page often visit “product B” next.
-  **3. Pattern Analysis (Knowledge Layer)**
- Evaluates, interprets, and visualizes discovered patterns.
- Removes irrelevant patterns and presents meaningful insights.
- Uses **OLAP tools, visualization, or statistical analysis.**
-  **Example:**
Showing graphs or reports of popular web pages and user navigation paths.
-  **Flow of Architecture**
- **Web Data → Data Extraction → Pattern Discovery → Pattern Analysis → Knowledge/Decision Making**